S^{2}HGC: An End-to-End Spectral–Spatial Hypergraph Convolutional Network for Unsupervised Hyperspectral Band Selection

Kaixiong Wu, Mingwei Wang, Haipeng Luo, Maolin Chen, Yujie Xu, and Wei Liu

***Abstract*— Band selection (BS) is an effective way to reduce redundancy and enhance processing efficiency by obtaining a subset with highly discriminative bands. However, existing methods focus on the correlations between neighboring bands, overlooking the nonlinear dependencies present in global contexts. Moreover, these approaches typically treat each band as an independent entity, disregarding the spatial structure and potential spectral–spatial information embedded within band combinations. To address these limitations, we propose S^{2}HGC, an end-to-end spectral–spatial hypergraph convolutional network designed for unsupervised hyperspectral BS. Our method eliminates error accumulation from stepwise processing and enables the holistic optimization of band subsets. Specifically, a controllable autoencoder (CAE) is designed to reflect the low-dimensional representation of full bands and partition them into multiple combinations. Furthermore, spatial topology is produced from these band combinations using superpixels, and a spectral–spatial information fusion (S^{2}IF) module is designed to extract spatial features and integrate them with spectral features. Finally, an adaptive hypergraph convolution is employed to capture the nonlinear correlations between bands and enhance the information interaction across long-distance bands. The critical bands are then selected to reconstruct the original hyperspectral imagery (HSI), which is iteratively optimized to obtain the optimal band subset. Extensive experiments on three publicly available datasets demonstrate the superior performance of S^{2}HGC compared to other state-of-the-art BS methods.**

***Index Terms*— Controllable autoencoder (CAE), hypergraph convolution, hyperspectral imagery (HSI), spectral–spatial information fusion (S^{2}IF), unsupervised band selection (BS).**

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I. Introduction

HYPERSPECTRAL imaging is a powerful technique capable of producing hundreds of contiguous narrow-band electromagnetic spectra, providing abundant information about different objects [1], [2]. This spectral information significantly improves the ability to recognize and classify complex or visually similar objects [3], [4], [5]. With rapid advancements in imaging technology, hyperspectral imagery (HSI) has garnered considerable attention and is now applied across various domains, including agriculture [6], [7], environmental monitoring [8], and mineral exploration [9], among others. Despite its advantages, HSIs exhibit strong correlations between neighboring bands due to their inherent characteristics [10], [11], [12]. The redundancy leads to challenges such as the Hughes phenomenon, also known as the "curse of dimensionality" [13]. These issues increase the computational and storage complexity of high-dimensional data and further complicate data processing and analysis. Consequently, effective dimensionality reduction is important for facilitating the utilization of HSIs.

For dimensionality reduction, feature extraction and feature selection are two primary techniques. Feature extraction projects the original high-dimensional data into a new low-dimensional feature space, with methods such as principal component analysis (PCA) [14], independent component analysis (ICA) [15], and linear discriminant analysis (LDA) [16], which has been applied in the HSI dataset. However, these approaches usually result in the loss of physical meaning of the original dataset in the transformed feature space [17]. In contrast, feature selection focuses on obtaining a subset of features as representativeness as possible directly from the high-dimensional data, preserving the inherent properties and physical meanings of the original dataset [18], [19], [20]. In the context of HSIs, feature selection is commonly referred to as band selection (BS), which plays a meaningful role in accomplishing subsequent interpretation and analysis of HSIs, making it a focal point of research in this field.

Based on the availability of labels, BS methods are categorized into supervised and unsupervised issues [21]. Supervised methods rely on labeled samples to train models for BS. However, acquiring sufficient labeled data is expensive and challenging, limiting the practicality of these methods in many real-world applications. In contrast, unsupervised methods do not require labeled samples, making them more widely

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applicable. These methods usually exploit prior knowledge acquired from the intrinsic geometric structure of raw HSIs to guide BS [22]. Unsupervised BS methods are broadly divided into three main strategies: ranking, searching, and clustering. Ranking strategies evaluate band importance using specific criteria, such as maximum variance [23], information entropy [24], [25], or structural similarity [26], to conduct BS. Searching strategies use heuristic or evolutionary algorithms to select discriminative bands [27], [28]. Clustering strategies group bands based on similarity and select crucial bands from each cluster [29]. However, traditional BS methods heavily rely on prior assumptions, making them dataset-specific with limited generalization capability. In addition, these methods primarily understand significant features, which restricts their ability to reflect enough feature representation.

In recent years, deep learning (DL) has achieved remarkable progress in HSI processing, outperforming traditional methods [30]. DL facilitates the automatic learning of deep and complex feature representation in an end-to-end manner, demonstrating superior generalization capability. As a result, deep neural networks (DNNs) have been extensively applied to conduct BS, with notable examples including convolutional neural networks (CNNs) [31], [32] and autoencoders (AEs) [33], [34]. Among them, CNNs, with their local receptive fields and multilayer nonlinear transformations, are particularly prevalent in BS applications [35]. Recent advances have begun to combine the robust feature extraction of CNNs with the adaptive decision-making capabilities of reinforcement learning, thereby enabling dynamic and efficient exploration of spectral characteristics [36]. In contrast, AE-based BS methods focus on efficiently reconstructing the latent characterization of high-dimensional data. These methods usually employ regularization or sparsity constraints to identify crucial bands. However, such constraints may fail to fully represent the complex data structure and feature diversity, resulting in challenges in managing sparse uncertainty. Moreover, due to architectural limitations, AE-based methods exhibit inadequate capabilities for modeling spatial information effectively.

HSIs contain both abundant spectral information and extensive spatial information. In recent years, spatial information has garnered significant attention, with superpixel-based methods being widely employed in BS [37]. For instance, the simple linear iterative clustering (SLIC) algorithm [38] segments an image into superpixel blocks that represent local spatial regions, preserving the spatial consistency of HSIs while reducing redundancy. Compared to BS methods that rely solely on spectral information, superpixel-based methods effectively delineate object boundaries and textures in HSIs by leveraging spatial information, thereby reducing the interference of redundant bands and noise in BS tasks. However, these methods segment superpixels from either the entire HSI or individual bands, without adequately considering the band variability across different objects. Furthermore, overemphasizing spatial information may lead to the neglect of spectral information in detail, making it challenging to effectively characterize spatial information in diverse band combinations.

To fully fuse spectral and spatial information, a graph structure is employed, where entities are represented as nodes and their relationships as edges. Among graph-based methods, graph convolutional networks (GCNs) [39] have gained popularity in BS due to their ability to effectively process non-Euclidean data. In BS applications, GCNs typically treat superpixels as nodes and their adjacency relationships as edges, enabling the extraction of spatial features through graph convolutions. This approach deepens the understanding of positional relationships among different objects. Additionally, GCNs model spectral information by representing bands as nodes and setting their correlations as edges, allowing the analysis of nonlinear relationships within the spectral domain. However, these methods address either spectral or spatial information in isolation, limiting their ability to fully integrate the strengths of both modalities. Furthermore, traditional GCN-based methods frequently overlook the information interaction across long-distance bands, resulting in insufficient performance of BS.

In this article, we propose an end-to-end spectral–spatial hypergraph convolutional (S^2HGC) network for unsupervised hyperspectral BS that consists of three main components: a controllable AE (CAE), a spectral–spatial information fusion (S^2IF) module, and an adaptive hypergraph convolution. These components are seamlessly integrated with an end-to-end training framework to enable holistic optimization and directly obtain the optimal band subset. First, the CAE reflects the low-dimensional representation of full bands, generating combinations of weakly correlated bands. Furthermore, the S^2IF module highlights the spatial structure of different band combinations within superpixels, fusing spectral–spatial information to define hypergraph nodes. Finally, an adaptive hypergraph convolution is employed to effectively capture the nonlinear correlations between bands. Additionally, the transition node is introduced to enhance the information sharing of discriminative bands. The main contributions of this article are summarized as follows.

1) A novel S^2HGC network is proposed for unsupervised hyperspectral BS, effectively reducing the dimensionality of the original HSI and directly obtaining the optimal band subset through an end-to-end training framework.
2) CAE and S^2IF modules are designed to generate combinations of weakly correlated bands and fuse spectral–spatial information across the entire HSI, thereby improving the efficiency and interpretability of BS.
3) An adaptive hypergraph convolution is constructed to capture the nonlinear correlations between bands, and the transition node is introduced to enhance the information interaction of bands and enrich band combinations.
4) Extensive experiments are conducted on three publicly available HSI datasets. The results demonstrate the superior performance of the proposed S^2HGC compared to other state-of-the-art BS methods.

The structure of this article is organized as follows. Section II provides a brief review of related work on BS. Section III details the proposed unsupervised BS method, S^2HGC. Section IV presents the experimental results on three publicly available datasets, demonstrating its effectiveness and

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superiority. Finally, Section V summarizes the main research findings and contributions of this study and outlines prospects for future research directions.

II. Related Work

In this section, we reviewed prior research related to the proposed S²HGC, including AE-based BS methods, superpixel-based BS methods, and graph-based BS methods. Moreover, the motivations behind this work are outlined to highlight the innovations and necessity of the proposed method.

A. AE-Based BS Methods

Regarding AE-based BS methods, extensive studies have focused on the properties of AEs to learn the hidden representation of HSIs and select crucial bands. Sun et al. [40] developed a concrete AE that employed the Gumbel–Softmax to form a discrete matrix, selecting the optimal band subset by maximizing information entropy, thus optimizing the BS process without relying on predefined assumptions to identify crucial bands (CAE-UBS). Tschannerl et al. [41] proposed an unsupervised and embedded BS method that utilized AE to reconstruct spectral features and identify crucial bands through sparsity constraints, thereby improving the quality-of-band subset (S-AEBS). Ahishali et al. [42] introduced a BS method that combined self-representation learning with a sparse 1-D operation autoencoder (SRL-SOA) to efficiently embody sparse representation, achieving sufficient learning ability even within a shallow architecture while mitigating overfitting. However, sparse constraints in AE-based methods are usually uncontrollable and may lead to the missing of critical bands or potential band combinations. Furthermore, most AE-based methods primarily focus on spectral information or directly flattening bands, neglecting the spatial structure inherent in HSIs.

B. Superpixel-Based BS Methods

Superpixel-based BS methods aim to learn the spatial structure of HSIs, thereby improving the ability of BS. Wang et al. [43] proposed a BS method based on region-aware latent feature fusion and clustering (RLFFC), which segmented the HSI into multiple regions using superpixels, and applied a fusion strategy to reflect shared feature representation for each region, enhance band separability, and reduce redundancy. Jia et al. [44] designed a multiscale superpixel-level group-clustering framework (MSGCF) that employed multiscale superpixel segmentation to form diverse superpixel maps, preserve the spatial information of HSIs and adapt to varying object scales, thereby improving the complementarity of BS. Subsequently, Tang et al. [45] proposed a BS method based on spatial and spectral structure-preserving self-representation (S⁴P), which utilized superpixels to extract the spatial features of bands, integrated spectral relevance and spatial similarity through a self-representation model and an adaptive weighted multiple graph fusion term, enabling effective unsupervised BS. However, these methods assess the importance of individual bands independently, without fully considering the spatial structure inherent in different band combinations leading to an insufficient expression of spatial information.

C. Graph-Based BS Methods

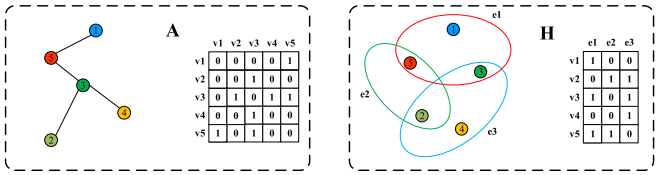
To address the limitations of CNNs, which are restricted to regular receptive fields, GCNs have emerged as a promising approach and have recently gained significant attention in the field of BS. Feng et al. [46] proposed a novel zero-order selection method (MR-Selection) that employed meta-reinforcement learning and a dynamic structure-aware GCN to efficiently capture the correlations between bands, and a two-stage optimization strategy was used to improve the efficiency and generalization capability. Cheng et al. [47] presented a BS method based on a dynamic-interactive subgraph Transformer (DISGT), which partitioned bands into dynamic subspaces, enhanced their information interaction to form subgraphs, and utilized a subgraph Transformer to comprehensively learn both local and global spectral information and accurately evaluate band importance. Zhang et al. [48] proposed a BS method leveraging dual-GCNs (BSD-GCNs), which encoded contextual and structural correlations between bands by constructing dual graphs based on bands and samples, incorporated band attention and sparsity constraints to improve the quality of BS. However, these methods mainly depend on the construction of adjacency matrices, where edges represent only pairwise relationships of attributes, making it challenging to uncover the potential and more suitable relational structures among bands.

D. Motivations

The aforementioned AE-based BS methods primarily focus on learning the spectral structure of HSI. However, they lack effective modeling of spatial information, which limits their ability to capture the correlations between bands for different objects and results in insufficient generalization capability. While superpixel-based BS methods preserve partial spatial information, they rely heavily on the segmentation of the entire HSI that overlooks the potential spatial textures inherent in band combinations and fails to account for information sharing of discriminative bands. Furthermore, graph-based methods typically represent only the pairwise relationship of attributes, which hinders their ability to reconstruct HSIs from multiple perspectives effectively.

As illustrated in Fig. 1, unlike traditional graph structure, hypergraph [49], [50] enables the connection of multiple nodes through a hyperedge, addressing the limitations of GCNs in modeling higher-order dependencies. Leveraging this aspect, S²HGC innovatively integrates hypergraphs into the BS process while capitalizing on the strengths of AEs and superpixels. By employing a CAE to learn the spectral structure, band combinations are delineated to gain object differentiation. The S²IF module is designed to effectively learn the potential spatial structure of these band combinations and facilitate complementary information fusion. Thereby, an adaptive hypergraph convolution is employed to highlight higher-order dependencies among bands, providing global

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(a) (b)

Fig. 1. Differences between a graph and a hypergraph. (a) Graph structure, where an edge connects exactly two nodes and is typically represented by an adjacency matrix A. (b) Hypergraph structure, where an edge connects any number of nodes and is usually represented by an incidence matrix H.

feature representation for BS. Ultimately, an end-to-end training framework allows for iterative optimization as a whole, enabling the direct obtaining of the optimal band subset.

III. Proposed Method

The architecture of the proposed S^{2}HGC is depicted in Fig. 2, which is primarily composed of four key modules: 1) a CAE for partitioning full bands; 2) an S^{2}IF for spectral–spatial information fusion; 3) an adaptive hypergraph convolution for selecting crucial bands; and 4) a reconstruction block for the evaluation of BS. A detailed explanation of the implementation processes and the specific functionality of each component is provided in the following.

A. Controllable Autoencoder

To effectively reduce redundancy in HSIs while preserving their spectral information, a CAE is employed to reflect the low-dimensional representation of the HSI dataset. By introducing a controllable parameter, the AE enables the partitioning of full bands. This mechanism determines an appropriate number of bands in each combination, thereby reducing computational complexity while maintaining the integrity of HSIs. The original HSI is represented as $C \in \mathbb{R}^{M\times N}$, where M denotes the number of samples and N represents the number of bands. A series of fully connected layers encoding C into a low-dimensional hidden representation $F \in \mathbb{R}^{M\times L}$, as shown in the following equation:

$$F = \sigma(W \cdot C + b) \tag{1}$$

where σ is the activation function, $W \in \mathbb{R}^{L\times N}$ is the weight matrix of encoder, and $b \in \mathbb{R}^L$ is the bias vector, $L \ll N$. L is set as the number of hidden layers, and it directly controls the number of band combinations by mapping the full bands into latent features. Assume that the weight matrix is $W = \{w_1, w_2, \ldots, w_L\}$, $w_i \in \mathbb{R}^N$ adaptively measures the correlations between bands and the ith low-dimensional feature representation. The index of the top K bands with the highest representation level is calculated by the absolute value of weight matrix $|w_i|$ and performing a descending sort, K is a controllable parameter to regulate the number of bands retained in each combination

$$o_K(w_i) = \left\{ j = \arg\max_{j'\in[1,N]} \mathcal{S}(A(w_i))^{(j')} \atop j' \notin o_{K-1}(w_i) \right\} \tag{2}$$

where $\mathcal{A}(w_i) = \{|w_i^{(1)}|, |w_i^{(2)}|, \ldots, |w_i^{(N)}|\}$, $w_i^{(j)}$ denotes the jth element of w_i. $\mathcal{S}(\mathcal{A}(w_i))$ represents the descending set of $\mathcal{A}(w_i)$, and $o_{K-1}(w_i)$ indicates the index set of the top $K-1$ largest elements, ensuring that the indices corresponding uniquely to the top K maximum. As shown in Fig. 3, where $K = 3$ is taken as an example to provide a detailed illustration of how the CAE partitions the full bands. In this manner, each w_i generates a band combination $o_K^{(i)}$ and the complete set of band combinations is given by $O = \{o_K^{(1)}, \ldots, o_K^{(i)}, \ldots, o_K^{(L)}\}$.

The band combination $o_K^{(i)}$ represents the K bands with the maximum representation when the full bands are mapped to the ith low-dimensional feature representation. Unlike traditional methods that rely on predefined rules for partitioning bands, CAE dynamically generates band combinations using a trainable weight matrix W. This allows for adaptive adjustment during an end-to-end training process. By partitioning full bands in HSIs into L band combinations, the proposed method improves the efficiency and interpretability of BS. This dynamic process also improves the usefulness of subsequent superpixel segmentation and hypergraph construction, contributing to a more effective BS process.

B. Spectral–Spatial Information Fusion

With the band combination $o_K^{(i)}$, fully associating these combinations with their corresponding objects becomes challenging. In principle, superpixels adaptively represent local spatial regions of a class, enabling the direct extraction of spatial features. The number of superpixels is proportionally increased to maintain adequate spatial granularity, too few superpixels lead to insufficient spatial feature extraction, and too many superpixels will reduce the processing efficiency. As a result, an appropriate number of superpixels is estimated by $P = 2^{\lfloor \lambda \cdot \log_{10} M + 2 \rfloor}$, where P denotes the number of superpixels and λ $(1.1 < \lambda < 1.5)$ is a scale factor to control the size of superpixels. As illustrated in Fig. 4, the S^{2}IF module is designed to learn the spatial information of various band combinations and perform a complementary fusion of spectral–spatial information, ensuring an overall representation of the HSI dataset.

To reduce the computational time while preserving spatial information, SLIC is applied to segment the HSI with $o_K^{(i)}$ into spatially connected superpixels. As shown in Fig. 5, each superpixel acts as a node and an undirected graph $g_i = (V, E)$ is constructed based on the neighborhood relationship between superpixels. The nodes V and edges E are represented by the superpixel neighborhood matrix $A^i \in \mathbb{R}^{P\times P}$ and the node feature matrix $h^i \in \mathbb{R}^{P\times d}$, where d denotes the dimension of features. The superpixel graph set is defined as $G = \{g_1, \ldots, g_i, \ldots, g_L\}$, and based on the neighborhood relationship between superpixels in the graph g_i, the adjacency matrix A^i is defined as follows:

$$A_{c,j}^i = \begin{cases} 1, & \text{if } S_c^i \text{ and } S_j^i \text{ are adjacent} \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where S_c^i denotes the cth superpixel of the ith band combination and consists of a collection of pixels within the region $\{p_1, p_2, \ldots, p_n\}$ and n denotes the number of pixels in S_c^i.

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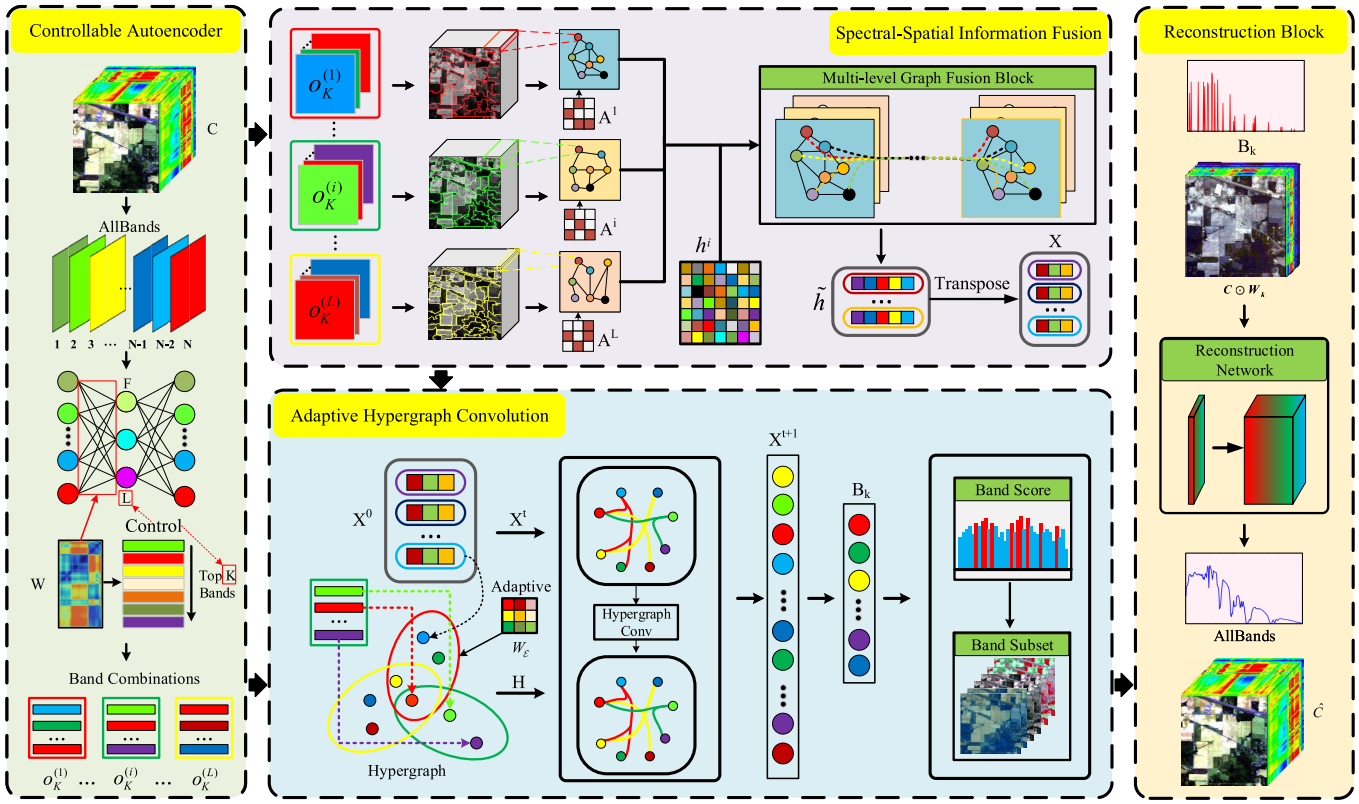


Fig. 2. Architecture of S²HGC.

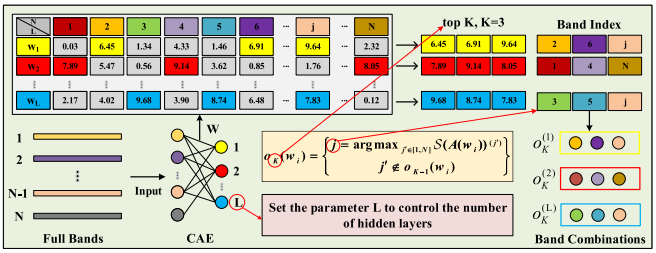


Fig. 3. Detailed illustration of CAE.

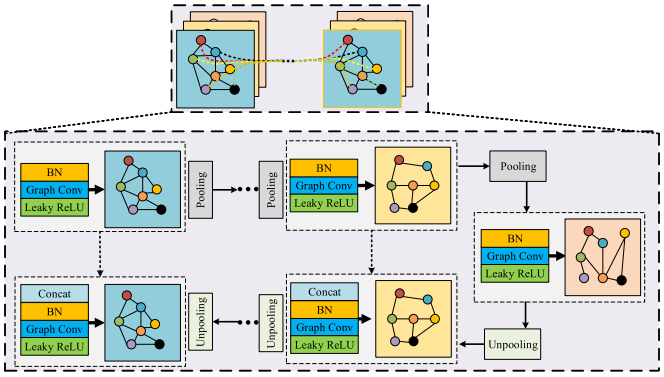
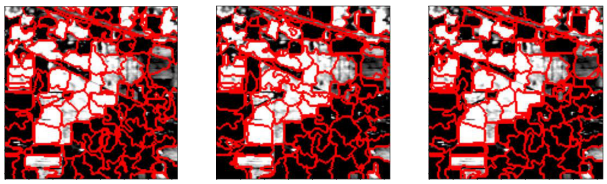


Fig. 4. Multilevel graph fusion block, which consists of a set of stacked attention graph convolutions and equal-number heterogeneous superpixel graph pooling functions to fuse spectral–spatial information.



(a) (b) (c)

Fig. 5. Using SLIC to segment HSIs with different band combinations on the Indian Pines dataset, with a fixed number of superpixels to 100. (a) Spatial regions of bands indexed 78, 63, 29, 130, 67. (b) Spatial regions of bands indexed 115, 66, 6, 100, 87. (c) Spatial regions of bands indexed 117, 53, 29, 130, 92.

Moreover, Jaccard similarity is utilized to reduce training time across consecutive iterations. A similarity threshold Q is defined; once the threshold is met, the superpixel graphs constructed based on current band combinations are similar to that of the previous iterations, enabling the reuse of the previously constructed superpixel graphs without reconstruction.

Following the graph construction, graph convolution is employed to produce the spatial topology. To excel in the extraction and aggregation of spatial information from neighboring nodes, an attention mechanism is applied to calculate edge weights based on the input features, allowing for adaptive adjustment of information aggregation. As a result, the attention-based graph convolution effectively learns the spatial structure of each band combination, represented as $\tilde{h^i}$

$$\tilde{h^i} = f_{\mathcal{G}}\left(h^i, \text{Sigmoid}\left(\left(h^i W\varphi\right)\left(h^i W\varphi\right)^{\mathrm{T}}\right) \odot A^i + \zeta I\right) \quad (4)$$

where $f_{\mathcal{G}}$ represents the graph convolution operation, W_φ and ζ are learnable parameters, and the operator $\odot$ denotes the

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Hadamard product. The identity matrix I is added for self-loops, enhancing the stability of the learning process.

To further explore the spatial structure of various band combinations within each superpixel and effectively integrate $\tilde{h}^i$, an equal-number heterogeneous superpixel graph pooling and unpooling function is employed to transform the spatial information of superpixels through an association matrix while maintaining a consistent number of superpixels for each band combination. Specifically, let $T^l \in \mathbb{R}^{P(l-1)\times P(l)}$, $(2 \le l \le L)$, represent the association matrix between $G(l-1)$ and $G(l)$, which is defined as follows:

$$T^l_{i,j} = \begin{cases} 1, & \text{if } \exists k \in [0, M-1] \\ & \quad Z_{l-1}(k) = i \text{ and } Z_l(k) = j \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

where Z represents the flattened superpixel label matrix. The equal-number heterogeneous superpixel graph pooling function from $G(l-1)$ to $G(l)$ is defined as $h^l = \text{Pool}(h^{l-1}) = (\hat{T}^l)^{\mathrm{T}} h^{l-1}$. The corresponding unpooling function is expressed as $h^{l-1} = \text{Unpool}(h^l) = T^l h^l$, where $\hat{T}^l$ denotes the column-normalized association matrix, calculated as $\hat{T}^l_{i,j} = T^l_{i,j} / \sum_m T^l_{m,j}$. Through these pooling and unpooling operations, the spatial information between different band combinations is propagated across multilevel graphs, thereby facilitating S^{2}IF. In addition, to embody pixel-level fine-grained feature extraction, an extra CNN layer is integrated into the encoding section to extract features from small-scale regular regions, extracting complementary spectral–spatial features at both pixel and superpixel levels.

C. Adaptive Hypergraph Convolution

Hypergraphs excel at highlighting higher-order dependencies among nodes, where a hyperedge connects multiple nodes. In the context of BS, bands are treated as nodes, enabling hypergraphs to effectively model the nonlinear correlations between bands. To explore potential higher-order dependencies among weakly correlated bands and enhance their information interaction, a hypergraph is constructed to treat the bands (incorporating spectral–spatial features) as hypergraph nodes, with each band combination corresponding to a hyperedge. The hypergraph is defined as $G = (\mathcal{V}, \mathcal{E}, W_{\mathcal{E}})$, where $\mathcal{V}$ is the set of vertices, representing the full bands, $\mathcal{E} = \{e_1, \ldots, e_i, \ldots, e_L\}$ is the set of hyperedges, corresponding to the band combinations O. Unlike traditional graphs, where an adjacency matrix A represents pairwise relationships between nodes, hypergraphs use an association matrix H to describe the relationships. Specifically, the rows and columns of A (size$|\mathcal{V}| \times |\mathcal{V}|$) denote vertices, while the rows and columns of H (size$|\mathcal{V}| \times |\mathcal{E}|$) represent the nodes and edges, respectively. The association matrix H is defined as follows:

$$H(v_i, e_j) = \begin{cases} 1, & \text{if } v_i \in e_j \\ 0, & \text{if } v_i \notin e_j \end{cases} \tag{6}$$

$$d(v_i) = \sum_{e_j \in \mathcal{E}} \omega(e_j) H(v_i, e_j) \tag{7}$$

$$d(e_i) = \sum_{v_i \in \mathcal{V}} H(v_i, e_j) \tag{8}$$

where $d(v_i)$ and $d(e_i)$ represent the node degree and edge degree, respectively, as computed from H, respectively. The feature set for each node (band) is defined as $X = \{x_1, x_2, \ldots, x_N\}, x_i \in \mathbb{R}^P$, and $X = \tilde{h}^T$. Due to variations in the information gained across different band combinations, assigning a uniform weight of one to each hyperedge is insufficient to fully highlight the higher-order dependencies among bands. To overcome this limitation, adaptive hyperedge weights are introduced. Specifically, a trainable weight parameter is assigned to each hyperedge, allowing the weights to dynamically adjust according to the information gained from band combinations. This adaptive mechanism prioritizes high-quality band combinations, while effectively mitigating the influence of redundant or less crucial bands. The hyperedge weights, $W_{\mathcal{E}}$, are expressed as $W_{\mathcal{E}} = \text{diag}(w^1_{\mathcal{E}}, \ldots, w^i_{\mathcal{E}}, \ldots, w^L_{\mathcal{E}})$, where each element $w^i_{\mathcal{E}}$ denotes the weight of the corresponding hyperedge e_i.

The hypergraph convolution utilizes H and X as inputs to perform node–hyperedge—node transformations via convolutional operations. This mechanism promotes the information flow between bands and effectively explores higher-order dependencies. Specifically, let $X^{(t)}$ and $X^{(t+1)}$ denote the input and output of the tth layer, respectively, with $Y^{(t)}$ being the hyperedge feature. The detailed formulation of the hypergraph convolution is provided in the following equations:

$$X^{(t+1)} = \sigma\left(D_v^{-1} H W_{\mathcal{E}} D_e^{-1} H^T X^{(l)} \Theta^{(l)}\right) \tag{9}$$

$$Y^{(t)} = W_{\mathcal{E}} D_e^{-1} H^T X^{(l)} \tag{10}$$

$$X^{(t+1)} = \sigma\left(D_v^{-1} H Y^{(t)} \Theta^{(l)}\right) \tag{11}$$

where σ is the RELU function for nonlinear activation. D_v and D_e denote the diagonal matrices of node degrees and edge degrees, mainly used to normalize the matrix H. Θ is a parameter to be learned during the training process. The hypergraph convolution consists of two stages. In the first stage, the node features $X^{(t)}$ are processed by the learnable parameter Θ, enabling continuous updates to the feature representation. These updated node features are then aggregated based on the hyperedges to compute the hyperedge features $Y^{(t)}$, as formulated in (10). In the second stage, the node features are further refined by aggregating the related hyperedge features, with the update process described in (11).

Hypergraph convolution incorporates a bi-directional information-passing mechanism, facilitating information interaction from nodes to hyperedges and vice versa. To further amplify the information sharing of discriminative bands, the transition node is introduced to enhance the information mobility within the hypergraph. The mathematical formulation of the transition node is defined as follows:

$$v_T = \lfloor \text{Sigmoid}(\alpha) \times N \rfloor \tag{12}$$

where α is a trainable parameter, the Sigmoid function maps α into the interval (0, 1), while $\lfloor \cdot \rfloor$ represents the floor function. As depicted in Fig. 6, the transition node v_T is connected to each hyperedge, acting as an information bridge. This allows the information flow between the nodes to extend beyond local hyperedge propagation, facilitating information interaction across long-distance bands with hyperedges. In all,

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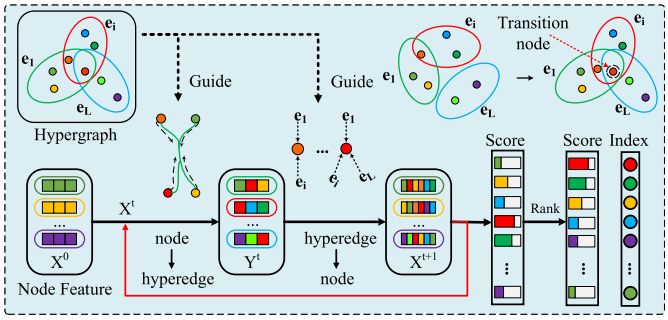


Fig. 6. Detailed illustration of adaptive hypergraph convolution and transition node.

the transition node enhances the information sharing of discriminative bands, further strengthening the ability to explore the dependencies among bands in the HSI dataset.

The nonlinear correlations between bands are effectively captured by hypergraph convolution. Building on this, a feature scoring rule is designed to quantify and assess the band importance, and the band score is calculated using the following equation:

$$s_i = \frac{1}{u}\sum_{i=1}^{u} X_i \tag{13}$$

where X_i denotes the features of the ith band and u is the dimension of features. Based on the scores $\{s_1, s_2, \ldots, s_N\}$, descending order is performed, and the k bands achieving the highest scores are selected as the optimal band subset with the following equation:

$$B_k = \arg\max_{B\subseteq\{1,2,\ldots,N\}} \sum_{i\in B} s_i \tag{14}$$
$$|B|=k$$

where B_k is the optimal band subset output by S²HGC.

D. Reconstruction Block

To assess the quality of the optimal band subset, a reconstruction operation is designed to evaluate the performance of the proposed S²HGC. Specifically, the weight matrix of band score $\mathcal{W}_k$, formed by hypergraph convolution, is applied to the original HSI. In this process, the weights corresponding to the unselected bands are set to 0, while the selected bands contribute to the reconstruction of the original HSI. This approach effectively isolates the impact of the selected bands, thereby evaluating their quality without interference from the remaining bands. The reconstruction operation is mathematically expressed as follows:

$$\hat{C} = \mathrm{Rec}(C \odot \mathcal{W}_k) \tag{15}$$

where $\hat{C}$ represents the reconstructed HSI. The weight matrix $\mathcal{W}_k$ corresponds to the selected k bands, with the weights determined by the band scores. The operation Rec is the reconstruction process and $\odot$ indicates element-wise multiplication. The reconstruction loss is evaluated using the mean squared error (mse) as the cost function, denoted as $\mathcal{L}$, which is defined as follows:

$$\mathcal{L} = \frac{1}{2M}\sum_{i=1}^{M} \|C_i - \hat{C}_i\|_2^2. \tag{16}$$

During the end-to-end training process, CAE, S²IF, adaptive hypergraph convolution, and reconstruction block are seamlessly integrated into the end-to-end training framework. Each module participates in holistic optimization, enabling the crucial bands to progressively reduce redundancy while retaining potential spectral–spatial information. The implementation details of S²HGC are presented in Algorithm 1.

Algorithm 1 S²HGC

Input: original HSI: C; size of band subset: k; number of superpixels: P; number of band combinations: L; number of bands per combination: K; number of training epochs: $\mathcal{I}$.

Output: Band subset: B_k

for 1,2,...,$\mathcal{I}$ **do**
  Partition full bands into combinations O by (1) and (2).
  for i=1,2,...,L **do**
    Segment $o_K^{(i)}$ into P superpixels by SLIC.
    Calculate the adjacency matrix A^i of the superpixel graph g_i by (3).
    Fuse spectral–spatial information h^i by (4) and (5).
  end for
  Calculate the association matrix H of hypergraph by (6).
  Update the node features X by (9)–(11).
  Calculate the band scores s by (13).
  Update the optimal band subset B_k by (14).
  Reconstruct the HSI $\hat{C}$ by (15).
  Calculate the reconstruction loss $\mathcal{L}$ by (16).
end for
return The top-ranked bands based on their scores are selected after model training to form the optimal band subset B_k.

IV. Experiments

In this section, experiments are conducted to validate the effectiveness of the proposed S²HGC. Section IV-A introduces three publicly available HSI datasets used for evaluation. Section IV-B outlines the experimental setup, including the comparison methods and evaluation metrics. Section IV-C presents a detailed analysis of the parameter sensitivity of S²HGC. In Section IV-D, a comparative analysis of the proposed method with existing approaches is performed, focusing on classification performance and the entropy values of selected bands. Section IV-E reports ablation experiments to further investigate the contribution of each component of the proposed method. Finally, Section IV-F compares the running times of different methods for selecting a specific number of bands.

A. HSI Datasets

The performance of the proposed method is validated on three public datasets: Indian Pines, Pavia University, and Dioni. The details of these datasets are as listed as Table I.

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TABLE I
SUMMARY OF INDIAN PINES, PAVIA UNIVERSITY, AND DIONI DATASETS

Datasets	Indian Pines	Pavia University	Dioni
Pixels	145 × 145	610 × 340	250 × 1376
Channels	200	103	176
Classes	16	9	12
Sensors	AVIRIS	ROSIS	Hyperion

TABLE II
HYPERPARAMETER SETTINGS FOR DIFFERENT BS METHODS

Comparative methods	Indian Pines	Pavia University	Dioni
BS-Net	λ=1e-2, η=2e-3, $maxiter$=100		
MH-DRL	α=10, γ=0.9, $step$=90		
CAE-UBS	lr=1e-3,B=512	lr=1e-3,B=8192	
S-AEBS	α=0.0001, β=0.01		
SRL-SOA	λ= 0.01,B=5,Q=5		
RLFFC	$\lambda = 2^{-15}$,β = 0.1,d=5		$\lambda = 2^{-15}$,β = 0.1,d=8
MSGCF	k=10	k=15	k=15
S^4P	α=100, γ=0.01, T=2000		
MR-Selection	α=-0.0001, β=0.01		
DISGT	S=3,∂ = 0.5,α= 0.1	S=2,∂ = 0.5,α= 0.1	S=3,∂ = 0.5,α= 0.1
BSD-GCN	η=1e-3, T=1000		

1) *Indian Pines:* Captured in 1992 by the AVIRIS sensor, this dataset consists of 224 bands spanning from 0.4 to 2.5 μm, with each band having a spatial resolution of 145 × 145 pixels. The dataset contains 16 land-cover classes. To remove bands that were severely affected by water absorption, 20 spectral bands (indices 104–108, 150–163, and 220) were discarded, leaving 200 bands for experiments.
2) *Pavia University:* Acquired in 2002 by the ROSIS sensor over Pavia, this dataset includes 115 bands with a spatial resolution of 1.3 m, which contains nine land-cover classes, with each band having a size of 610 × 340 pixels. Twelve noisy bands were removed, resulting in 103 bands used for further analysis.
3) *Dioni:* This dataset, collected from the HyRANK benchmark, was collected by the Hyperion sensor on the Earth Observing-1 (EO-1) satellite operated by NASA. It consists of 176 bands, each with a spatial resolution of 250 × 1376 pixels, and includes 12 land cover classes.

B. Experimental Settings

1) Comparison Methods: To validate the effectiveness of the proposed S^{2}HGC, a comparison is conducted with 11 other state-of-the-art BS methods. These methods include CNN-based methods: BS-Net [35]. DRL-based methods: MH-DRL [36]. AE-based methods: CAE-UBS [40], S-AEBS [41], and SRL-SOA [42]. Superpixel-based methods: RLFFC [43], MSGCF [44], and S^4P [45]. Graph-based methods: MR-Selection [46], DISGT [47], and BSD-GCN [48]. Moreover, AllBands is used as a baseline reference, where the full bands of the HSI dataset are utilized. Each method was executed 20 times to ensure the reliability and consistency of the results. Table II lists the hyperparameter settings of all comparison methods.

2) Evaluation Metrics: The quality of the selected bands is assessed based on the classification performance using three typical classifiers: support vector machines (SVMs) [51] with a radial basis function, k-nearest neighbors (kNNs) [52] with $k = 5$, and random forest (RF) [53] with 200 trees. For each dataset, 10% of the samples from each class are randomly selected for training, while the remaining 90% are used for testing. To minimize the effect of random sampling bias, each classifier is executed 20 times, and the average classification results are reported. Three commonly used metrics are employed for evaluation: overall accuracy (OA), average accuracy (AA), and Kappa coefficient (Kappa).

3) Environment Configuration: First, all the experiments are conducted on a system running Windows 10, with an Intel Xeon E-2286M CPU, an Nvidia Quadro RTX 5000 GPU, 32 GB of RAM, and 16 GB of GPU memory. The software environment consists of Python 3.6 and PyTorch 1.9.1 + cu111. In all experiments, Adam is used as the optimizer and a step decay strategy is adopted to ensure stable convergence. Specifically, the learning rate is initialized to 0.001 and is reduced by a factor of 0.5 for 20 epochs. All models are trained 100 epochs for the Indian Pines, Pavia University, and Dioni datasets with a learning rate of 0.001.

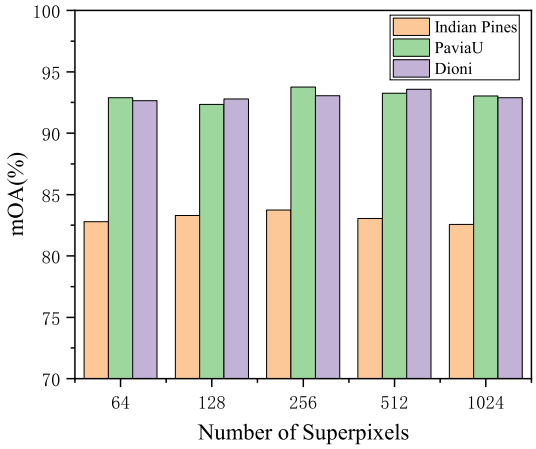


Fig. 7. mOA for different numbers of superpixels on three HSI datasets.

C. Parametric Analysis

Sensitivity analysis is conducted on the parameters for selecting crucial bands through fine-tuning. Among them, S^{2}HGC has four key parameters: the number of band combinations L, the number of bands in each combination K, the number of superpixels P, and the similarity threshold Q. As shown in Table III, Q has minimal impact on the mean OA (mOA) across different datasets, with a slight improvement in OA as Q increases. However, its effect on training time is more pronounced. Specifically, when Q is set to 80%, the training time on the Indian Pines dataset is reduced to nearly 80% compared to when Q is not used. The impact is even more significant for the Pavia University and Dioni datasets, where training time is reduced by approximately tenfold. By balancing OA and training time, Q is set as 80% for S^{2}HGC.

Determining the number of superpixels for different datasets is challenging in practice. In this study, we investigate the impact of varying the number of superpixels by considering a range of values: 64, 128, 256, 512, and 1024, as shown in Fig. 7. The results indicate that S^{2}HGC is not highly sensitive to the number of superpixels. However, both an excessively

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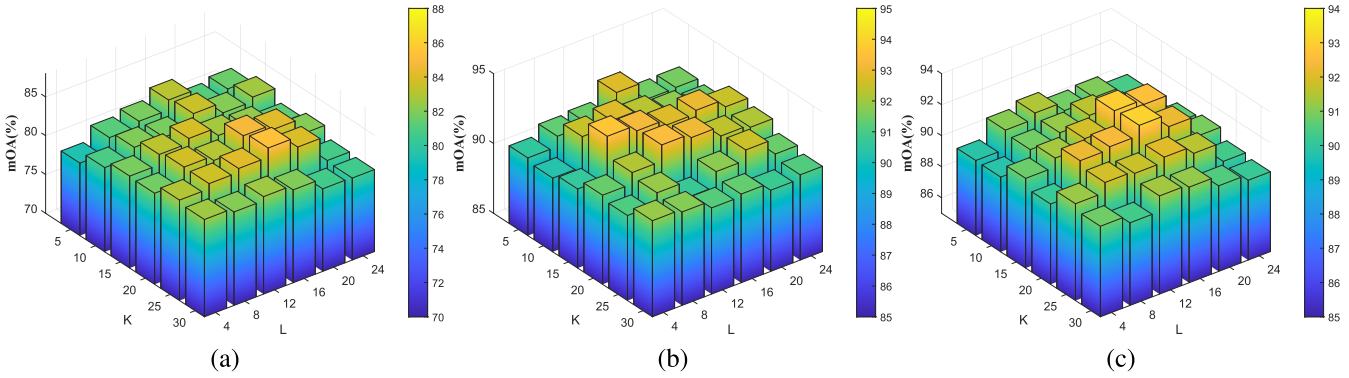


Fig. 8. Sensitivity analysis of parameters on three datasets: (a) Indian Pines, (b) Pavia University, and (c) Dioni.

TABLE III
COMPARISON OF RUNNING TIME AND MOA RANGING FROM 5 TO 30 BANDS FOR DIFFERENT Q VALUES

Datasets	Q(%)	RUNNING TIME (s)	mOA(%)
Indian pines	70	95.68	82.96
	80	167.63	83.83
	90	228.16	83.65
	100	564.96	83.73
	non-use	736.18	83.79
Pavia University	70	286.13	93.66
	80	416.02	93.75
	90	888.89	93.43
	100	2569.11	93.81
	non-use	5265.42	93.78
Dioni	70	345.31	92.79
	80	612.53	93.09
	90	947.67	93.12
	100	3053.86	93.04
	non-use	5938.37	93.23

high or low number of superpixels may lead to a slight decrease in OA. A large number of superpixels will cause an uneven distribution of spatial information, which results in insufficient feature representation for each superpixel. Furthermore, too few superpixels may lead to the merging of different objects or regions, blurring the spatial structure and impairing the accurate recognition of object boundaries and shapes. Specifically, the number of superpixels is set as 256 for the Indian Pines and Pavia University datasets, and 512 for the Dioni dataset.

Regarding the parameters L and K, experiments are conducted by fixing one parameter while varying the other. First, $L = 4$ is fixed while K is varied. Then, $K = 5$ is fixed while L is varied. Fig. 8 illustrates the mOA as it ranges from 5 to 30 bands at different values across three datasets. The results indicate that when L is fixed, increasing K initially leads to an improvement in mOA, followed by a subsequent decline. A similar trend is observed when K is fixed, and L is increased. Overall, the mOA rises with increasing values of L and K until it reaches a peak, after which it begins to decline, forming a "mountain" shape with a higher center and lower surrounding areas. The trend is attributed to the fact that both L and K directly influence the range of information interaction: excessively high values of L or K may cause redundant and noisy information, while too few hyperedges or nodes result in the loss of critical bands, failing to adequately capture the correlations between bands. Therefore, selecting appropriate values for L and K is essential for obtaining the optimal band subset. Specifically, the reasonable parameters are found to be $L = 16$, $K = 25$ for the Indian Pines dataset, $L = 8$, $K = 15$ for the Pavia University dataset, and $L = 16$, $K = 20$ for the Dioni dataset. When $L = 16$ and $K = 20$, S²HGC demonstrates strong classification performance across all three datasets.

D. Experimental Results

In this section, the performance of S²HGC is evaluated on the Indian Pines, Pavia University, and Dioni datasets. A detailed comparative analysis is conducted between S²HGC and other state-of-the-art BS methods. This analysis focuses on several aspects, including classification performance, distribution of band subsets, and entropy value of bands.

1) Results on the Indian Pines Dataset: Fig. 9(a)–(c) presents the OA of different BS methods and the baseline AllBands using SVM, kNN, and RF classifiers. The results indicate that the proposed S²HGC consistently achieves higher OA across three classifiers in most scenarios. For the SVM classifier, when the number of selected bands exceeds 15, S²HGC significantly outperforms other methods. Remarkably, when more than 20 bands are selected, the OA of S²HGC surpasses that of the baseline, underscoring its superior BS performance. In contrast, other methods require a larger number of bands to retain classification performance comparable to the baseline. Furthermore, Fig. 9(a) demonstrates that increasing the number of selected bands does not always improve OA. For instance, when more than 25 bands are selected, methods such as BS-Net, MSGCF, and MR-Selection exhibit a decline in OA. In contrast, S²HGC effectively continues to identify crucial bands, mitigating the Hughes phenomenon and maintaining robust performance.

To provide a quantitative analysis of the performance of different methods, Table IV presents a detailed comparison of OA using SVMs to select 25 bands. The OA of S²HGC is higher than other methods across ten classes. Moreover, S²HGC outperforms others in terms of AA and Kappa coefficient, further demonstrating its effectiveness. Fig. 10 illustrates the distribution of band subsets obtained by different methods, along with the corresponding entropy value for each band. Compared to methods such as CAE-UBS, S-AEBS, SRL-SOA, RLFFC,

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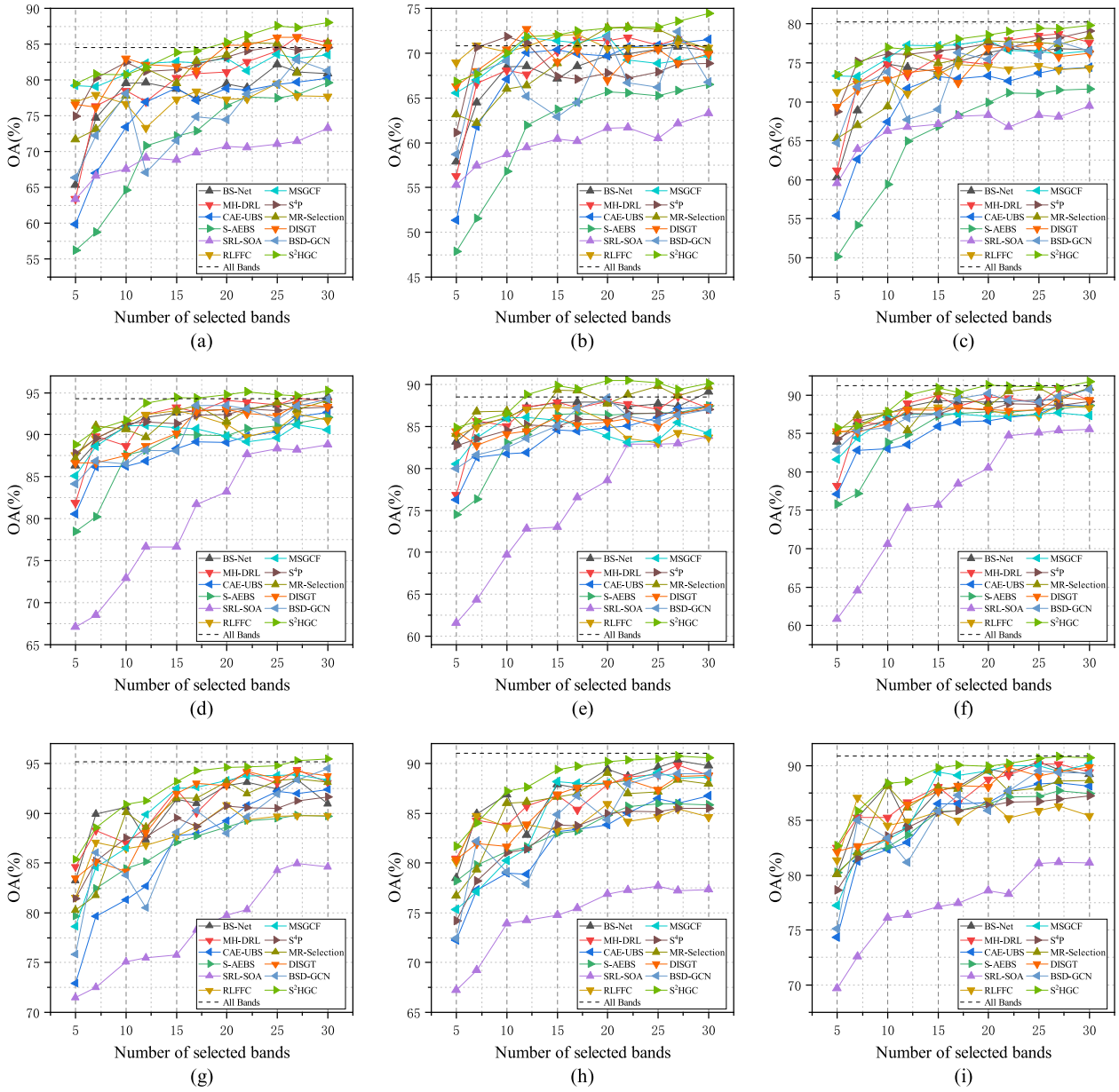


Fig. 9. Classification performance of 11 BS methods under a different number of selected bands. (a)–(c) Classification accuracy on the Indian Pines dataset using SVM, kNN, and RF as classifiers. (d)–(f) Classification accuracy on the Pavia University dataset using SVM, kNN, and RF as classifiers. (g)–(i) Classification accuracy on the Dioni dataset using SVM, kNN, and RF as classifiers.

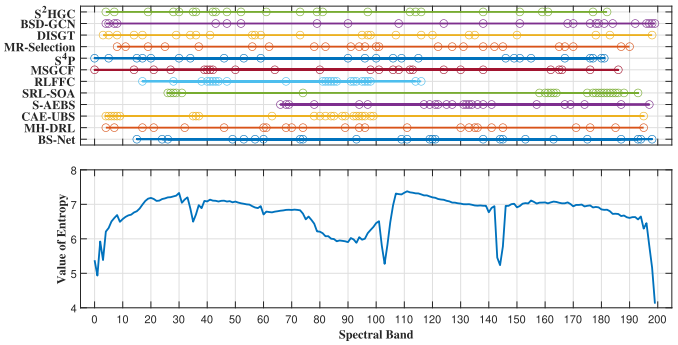


Fig. 10. Distribution of 25 bands selected by different BS methods on the Indian Pines dataset and the entropy value for each band.

and BSD-GCN, S^{2}HGC exhibits a more uniform distribution of band subset. This uniformity enables S^{2}HGC to effectively avoid selecting low-entropy bands, indicating its ability to identify crucial bands that contribute significantly to HSIs. Fig. 11 showcases the classification maps generated by different BS methods using SVMs to select 25 bands. Among these methods, the classification map produced by S^{2}HGC is notably clearer and more distinct. This demonstrates the effectiveness of S^{2}HGC in selecting high-quality bands, which are better suited for subsequent HSI classification tasks.

2) Results on the Pavia University Dataset: Fig. 9(d)–(f) depicts the OA achieved by various BS methods using SVM, kNN, and RF classifiers. The results reveal that, except for SRL-SOA, S-AEBS, and CAE-UBS, other methods follow similar classification performance. This behavior may be attributed to the characteristics of the dataset, which includes diverse and separable objects and consists

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TABLE IV

CLASSIFICATION ACCURACY OF DIFFERENT BS METHODS USING 25 BANDS ON THE INDIAN PINES DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

No.	BS-Net	MH-DRL	CAE-UBS	S-AEBS	SRL-SOA	RLFFC	MSGCF	S^{4}P	MR-Selection	DISGT	BSD-GCN	S^{2}HGC
1	82.08±11.91	76.52±8.54	76.04±10.31	71.92±8.32	56.77±12.70	52.98±6.21	91.44±5.82	81.99±7.75	63.39±11.82	80.73±5.39	72.24±15.90	**92.40±4.71**
2	73.14±0.32	78.68±0.44	66.61±0.36	71.95±2.40	60.78±2.91	65.51±1.22	79.66±2.45	78.25±1.51	78.05±0.99	79.18±0.73	70.60±0.76	**80.66±1.79**
3	74.40±0.60	77.18±2.62	67.56±1.82	79.28±3.80	58.86±2.27	74.42±2.69	80.37±1.18	78.37±2.71	77.87±0.58	80.71±2.17	71.79±1.00	**82.92±1.32**
4	63.78±2.85	69.59±0.77	64.31±1.77	61.89±2.57	61.61±2.71	54.86±5.08	73.55±2.05	**78.55±1.52**	69.01±6.06	74.59±3.82	63.63±3.62	74.57±1.37
5	90.50±0.91	87.92±1.95	86.83±1.84	87.71±3.51	78.02±0.90	88.15±0.93	89.92±3.13	**91.20±3.10**	87.54±0.99	90.81±3.57	83.00±2.59	90.98±1.67
6	89.75±0.42	89.00±1.08	92.86±0.97	88.87±2.06	82.60±1.36	90.22±0.84	91.51±0.33	92.91±0.74	91.72±0.50	92.81±0.76	88.50±2.29	**95.51±0.36**
7	78.55±12.03	94.32±6.88	86.32±6.03	68.30±19.66	83.88±7.38	85.12±3.20	90.19±4.64	85.44±12.67	95.06±4.28	94.89±5.17	65.28±10.49	**98.72±2.22**
8	94.22±0.39	95.71±1.41	94.63±0.66	92.58±1.75	92.42±1.12	93.73±2.91	96.75±1.20	94.68±0.84	93.95±1.98	**97.26±2.01**	94.82±0.80	96.90±2.00
9	65.54±12.84	76.87±15.94	66.82±10.62	58.19±5.96	60.10±8.88	71.88±14.81	68.69±14.58	82.54±3.61	65.25±11.89	71.52±4.90	65.17±21.77	**86.09±12.08**
10	81.41±1.58	82.07±0.94	74.94±1.38	67.08±3.14	59.02±0.96	79.53±0.77	80.40±1.50	82.58±1.77	82.89±0.45	80.53±1.51	75.28±0.48	**83.42±1.20**
11	80.85±0.24	83.53±1.16	79.71±0.60	68.30±0.61	68.10±0.90	78.82±2.52	83.62±1.27	83.68±0.80	84.53±0.86	86.00±0.48	78.38±2.17	**87.43±1.46**
12	80.84±1.39	84.00±0.50	78.35±4.73	78.83±4.33	61.18±0.97	78.90±2.63	**90.07±0.65**	77.63±2.77	89.50±0.57	85.69±1.45	81.96±2.65	85.10±1.23
13	93.24±1.91	91.84±3.01	90.76±3.73	94.42±1.95	80.03±2.42	92.98±1.29	94.19±1.01	92.44±2.28	**96.40±0.48**	92.98±0.91	92.12±2.25	94.15±1.81
14	92.23±0.95	93.15±0.87	93.38±1.49	93.27±0.83	89.36±1.56	92.86±0.50	93.00±0.63	93.43±0.44	94.73±0.35	94.34±0.59	90.61±0.87	**96.42±0.54**
15	73.13±2.48	69.29±4.06	67.70±2.52	73.44±4.29	63.04±1.74	70.56±5.16	76.20±4.59	74.60±1.69	74.72±4.69	76.42±1.55	67.06±1.13	**79.97±1.56**
16	99.22±0.68	98.40±0.73	96.96±3.20	93.57±3.09	98.90±1.90	98.01±0.64	**99.25±0.65**	98.52±0.62	96.31±3.44	98.83±0.03	98.13±2.30	98.44±0.64
OA(%)	82.18±0.27	84.08±0.38	79.35±0.31	77.48±0.54	71.03±0.56	79.55±0.21	85.11±0.31	84.68±0.24	85.09±0.13	85.95±0.24	79.46±0.37	**87.55±0.29**
AA(%)	82.05±0.92	84.25±0.41	80.24±1.47	78.10±1.63	72.17±0.20	79.28±0.57	86.18±1.15	85.42±0.60	83.81±0.46	86.08±0.86	78.66±2.20	**88.98±0.71**
Kappa(%)	79.64±0.30	81.83±0.45	76.45±0.37	74.11±0.60	66.77±0.61	76.66±0.23	83.01±0.37	82.50±0.28	83.00±0.14	83.98±0.28	76.56±0.45	**85.80±0.35**

TABLE V

CLASSIFICATION ACCURACY OF DIFFERENT BS METHODS USING 20 BANDS ON THE PAVIA UNIVERSITY DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

No.	BS-Net	MH-DRL	CAE-UBS	S-AEBS	SRL-SOA	RLFFC	MSGCF	S^{4}P	MR-Selection	DISGT	BSD-GCN	S^{2}HGC
1	92.81±0.34	93.10±0.12	92.26±0.35	91.64±0.76	84.87±2.32	92.82±0.18	93.13±0.85	93.24±0.24	91.51±0.18	93.43±0.37	92.91±0.37	**94.32±0.17**
2	95.76±0.20	96.98±0.10	92.41±0.10	91.97±0.17	86.85±0.35	92.79±0.08	89.97±0.32	94.68±0.20	94.92±0.17	94.96±0.15	95.75±0.05	**97.13±0.18**
3	82.10±0.49	84.18±1.13	75.99±0.99	78.06±0.69	62.64±3.52	80.39±0.42	81.10±0.58	82.60±0.68	71.78±1.15	**85.69±1.43**	85.31±1.58	85.52±1.55
4	97.36±0.34	97.00±0.39	95.15±0.67	94.93±0.58	91.33±0.41	95.34±0.87	95.34±0.65	95.73±0.27	96.67±0.38	95.65±0.47	95.97±0.70	**97.68±0.58**
5	99.68±0.17	99.75±0.04	99.63±0.15	99.65±0.16	99.53±0.01	99.82±0.04	99.23±0.11	99.58±0.42	99.78±0.15	99.78±0.15	99.70±0.20	**99.83±0.17**
6	90.17±0.55	92.73±0.31	85.32±0.20	86.64±0.28	79.40±0.23	87.96±0.63	84.97±1.08	89.33±0.34	91.57±0.57	89.99±0.90	91.29±0.68	**94.20±0.39**
7	82.92±0.97	85.22±1.21	79.10±0.65	78.98±1.34	59.47±1.31	82.96±0.33	85.56±0.71	84.88±2.15	76.67±2.34	85.44±2.15	85.28±3.35	**85.84±0.98**
8	84.52±0.39	86.06±0.37	80.89±1.44	80.23±1.02	66.67±1.12	84.53±0.31	84.37±1.14	83.93±0.45	80.42±0.16	86.18±0.74	86.96±0.87	**87.35±0.71**
9	99.96±0.06	**100.00±0.00**	99.96±0.06	99.89±0.00	99.54±0.34	**100±0.00**	**100±0.00**	99.96±0.06	99.89±0.00	99.96±0.06	99.93±0.06	99.89±0.00
OA(%)	92.99±0.02	94.12±0.06	90.02±0.02	89.91±0.09	83.20±0.31	91.26±0.08	89.85±0.10	92.39±0.06	91.52±0.01	92.98±0.13	93.46±0.07	**94.78±0.14**
AA(%)	91.70±0.06	92.78±0.19	88.97±0.11	89.11±0.20	81.14±0.11	90.73±0.11	90.41±0.12	91.55±0.28	89.25±0.07	92.34±0.37	92.57±0.32	**93.53±0.21**
Kappa(%)	90.68±0.03	92.20±0.08	86.67±0.03	86.50±0.13	77.24±0.44	88.32±0.10	86.37±0.15	89.87±0.07	88.71±0.02	90.66±0.17	91.30±0.10	**93.07±0.18**

TABLE VI

CLASSIFICATION ACCURACY OF DIFFERENT BS METHODS USING 27 BANDS ON THE DIONI DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

No.	BS-Net	MH-DRL	CAE-UBS	S-AEBS	SRL-SOA	RLFFC	MSGCF	S^{4}P	MR-Selection	DISGT	BSD-GCN	S^{2}HGC
1	82.33±1.31	87.16±0.62	75.52±2.99	74.41±1.38	68.43±1.00	69.70±2.03	84.73±1.04	80.65±0.73	83.91±0.38	86.45±0.33	82.87±0.08	**90.38±0.46**
2	91.76±1.98	91.09±3.91	93.88±0.86	88.37±2.09	88.67±2.34	95.72±1.68	95.87±1.43	91.65±2.92	**95.90±1.21**	95.60±1.43	92.25±1.71	94.69±1.15
3	92.81±0.20	90.86±1.33	88.01±0.42	82.38±2.19	80.71±1.97	83.60±2.15	91.30±0.86	86.68±1.29	91.84±1.20	90.45±1.82	90.80±1.34	**93.37±2.16**
4	84.72±4.30	86.25±2.98	88.06±3.19	76.79±3.73	46.43±0.24	92.50±3.33	92.41±1.99	83.61±3.06	85.92±2.19	88.38±5.14	81.99±3.53	**92.87±3.16**
5	92.97±1.28	94.17±0.41	88.83±1.15	85.17±0.86	79.71±1.67	88.37±1.37	93.72±0.32	90.45±1.32	92.86±1.49	93.02±1.10	92.94±0.51	**94.60±0.21**
6	96.67±1.18	98.51±0.69	97.28±1.60	96.53±1.20	82.58±4.61	98.27±1.83	98.54±0.41	94.64±2.58	98.70±0.65	**99.08±0.57**	97.60±0.43	97.39±0.66
7	95.10±0.55	95.78±0.59	95.78±0.49	94.57±0.40	91.43±0.22	95.60±0.29	95.79±0.23	94.14±0.68	96.03±0.62	95.41±0.64	95.23±0.53	**96.06±0.07**
8	92.74±0.45	93.67±0.40	91.64±0.19	88.95±0.70	84.71±0.42	90.29±0.12	92.82±0.15	91.72±0.45	92.03±0.67	93.24±0.28	92.83±0.46	**94.22±0.34**
9	93.62±1.23	93.20±0.73	88.66±1.14	86.35±0.27	82.07±1.01	87.44±0.99	91.33±1.47	90.33±1.11	91.31±1.82	94.94±0.79	91.59±1.77	**95.91±0.47**
10	95.39±2.77	97.61±0.59	96.77±0.84	96.33±0.35	92.03±1.38	96.43±1.35	98.14±0.52	**98.56±1.07**	96.66±1.10	97.72±1.27	97.57±1.03	98.44±0.42
11	99.26±0.16	99.63±0.33	99.69±0.06	98.97±0.13	91.49±0.49	88.96±0.40	99.92±0.04	92.86±0.17	**99.92±0.09**	99.79±0.22	99.13±0.35	99.69±0.22
12	97.89±0.73	99.50±0.65	99.75±0.00	98.63±0.97	78.83±1.09	84.12±0.96	99.46±0.36	79.47±1.53	99.42±0.52	99.33±0.38	98.14±2.33	**99.92±0.15**
OA(%)	93.41±0.34	94.39±0.11	91.97±0.08	89.82±0.23	84.93±0.23	89.76±0.07	93.86±0.04	91.22±0.18	93.48±0.27	94.23±0.11	93.33±0.42	**95.29±0.14**
AA(%)	92.94±0.81	93.95±0.23	91.99±0.33	88.95±0.26	80.59±0.60	89.25±0.06	94.50±0.21	89.56±0.35	93.71±0.15	94.45±0.63	92.75±0.64	**95.63±0.07**
Kappa(%)	91.83±0.43	93.05±0.13	90.04±0.09	87.36±0.30	81.22±0.29	87.27±0.09	92.38±0.05	89.12±0.22	91.91±0.33	92.84±0.14	91.73±0.51	**94.16±0.17**

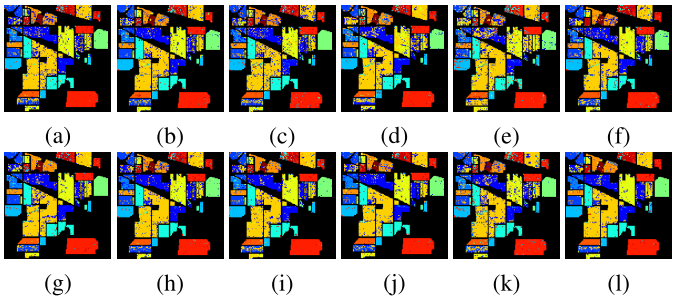


Fig. 11. Classification maps of different BS methods on the Indian Pines dataset with 25 bands by using the SVM: (a) BS-Net, (b) MH-DRL, (c) CAE-UBS, (d) S-AEBS, (e) SRL-SOA, (f) RLFFC, (g) MSGCF, (h) S^{4}P, (i) MR-Selection, (j) DISGT, (k) BSD-GCN, and (l) S^{2}HGC.

of only 103 bands, enabling discriminative bands to overall characterize the dataset. In this context, the proposed S^{2}HGC consistently delivers relatively higher OA compared to other methods. More importantly, it surpasses the baseline across three classifiers, underscoring its robust ability to effectively identify crucial bands that improve classification performance.

Table V presents a comprehensive comparison of OA across different methods when selecting 20 bands. The proposed S^{2}HGC demonstrates superior performance, with the highest OA in seven classes and significantly enhancing AA and Kappa coefficient compared to other methods. Fig. 12 illustrates the distribution of band subsets and the corresponding entropy value for each band. The entropy curve of the Pavia University dataset is relatively smooth, without regions of sharp decline. The bands selected by S^{2}HGC exhibit higher entropy values and are more widely distributed, avoiding the concentration and redundant information observed in the bands selected by methods such as MR-Selection, MSGCF,

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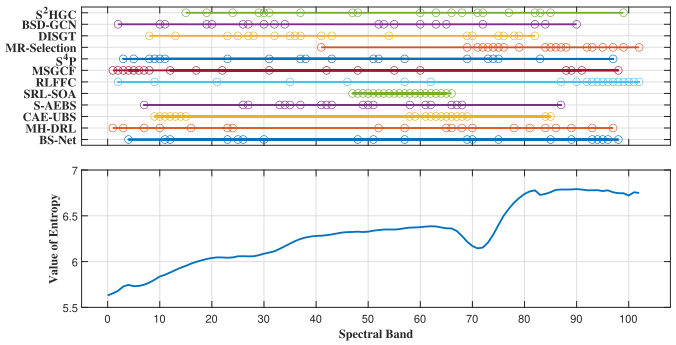


Fig. 12. Distribution of 20 bands selected by different BS methods on the Pavia University dataset and the entropy value for each band.

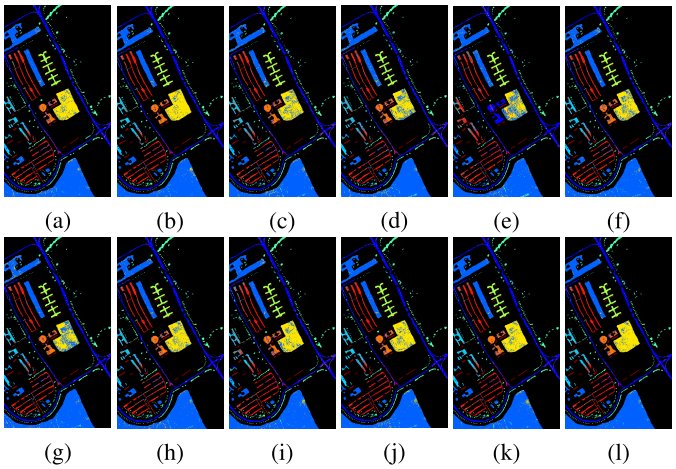


Fig. 13. Classification maps of different BS methods on the Pavia University dataset with 20 bands by using the SVM: (a) BS-Net, (b) MH-DRL, (c) CAE-UBS, (d) S-AEBS, (e) SRL-SOA, (f) RLFFC, (g) MSGCF, (h) S^{4}P, (i) MR-Selection, (j) DISGT, (k) BSD-GCN, and (l) S^{2}HGC.

RLFFC, SRL-SOA, and CAE-UBS. This dispersed distribution of high-entropy bands contributes to improving classification performance. Fig. 13 showcases classification maps generated by different BS methods using SVM with 20 selected bands. The results clearly show that the bands selected by S^{2}HGC yield more accurate and visually clear classification maps across most classes. This improvement is attributed to the crucial bands selected by S^{2}HGC, which effectively minimizes noise and enhances the fusion of potential spectral–spatial information.

3) Results on the Dioni Dataset: As illustrated in Fig. 9(g)–(f), the OA achieved by different BS methods using SVM, kNN, and RF classifiers shows that S^{2}HGC consistently improves the classification performance as the number of selected bands increases across three classifiers. In contrast, other methods, such as MH-DRL, MR-Selection, and BSD-GCN, exhibit inconsistent trends, with OA either plateauing or declining in certain cases. This decline suggests that these methods may include redundant bands that lead to unnecessary features. Notably, when the number of selected bands exceeds 27, S^{2}HGC maintains and even surpasses the OA of the baseline. This demonstrates its robustness in identifying crucial bands and achieving superior performance in BS tasks.

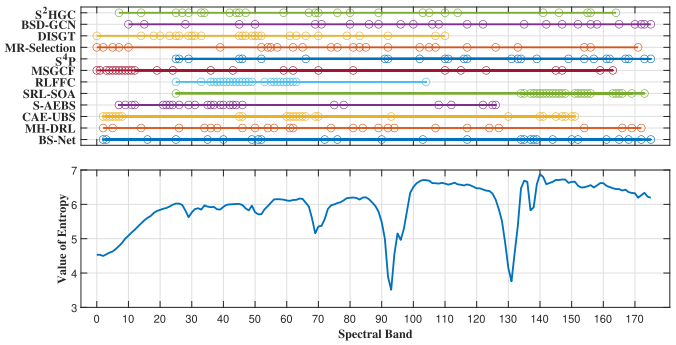


Fig. 14. Distribution of 27 bands selected by different BS methods on the Dioni dataset and the entropy value for each band.

To further evaluate the optimal band subset, a comprehensive assessment was conducted on the 27 bands selected by different BS methods. Table VI details the specific OA across 12 classes. The results reveal that S^{2}HGC achieves the highest OA in eight classes compared to other methods, alongside markedly superior AA and Kappa coefficient, underscoring its effectiveness in fusing spectral–spatial information. Fig. 14 visually compares the distribution of band subsets and the entropy value of each band. The bands selected by S^{2}HGC exhibit a more uniform distribution and avoid regions with sharp entropy declines. This proves the ability to emphasize crucial bands while minimizing redundancy, effectively learning relevant information to improve classification performance. The classification maps in Fig. 15 further validate the effectiveness of S^{2}HGC. These maps demonstrate that S^{2}HGC consistently produces clearer delineations of challenging-to-distinguish objects, preserving boundaries between classes even in areas with significant spectral overlap, improving classification performance while maintaining high spatial fidelity.

E. Ablation Experiments

During the evaluation of S^{2}HGC, two variants were introduced for comparative analysis: HGCBS, which excludes the S^{2}IF module, and S^{2}HGC-Random, which employs random partitioning for band combinations in the first stage. By comparing the performance of S^{2}HGC with these variants, valuable insights into the contributions of each module to classification performance are derived. Table VII summarizes the mOA of S^{2}HGC and its variants when using SVM across three datasets. The results clearly demonstrate that S^{2}HGC consistently outperforms the variants, indicating the important role of its integrated design. Notably, on the Indian Pines dataset, the mOA of S^{2}HGC surpasses that of HGCBS by 4.19%, underscoring the significance of incorporating spatial information for BS and validating the effectiveness of the S^{2}IF module in improving classification performance.

Moreover, S^{2}HGC demonstrates substantial performance improvements over S^{2}HGC-Random across three datasets, affirming the reliability and effectiveness of the CAE module compared to random partitioning strategies. The end-to-end training framework promotes the synergy among independent modules and significantly improves learning efficiency, enabling the model to capture the nonlinear

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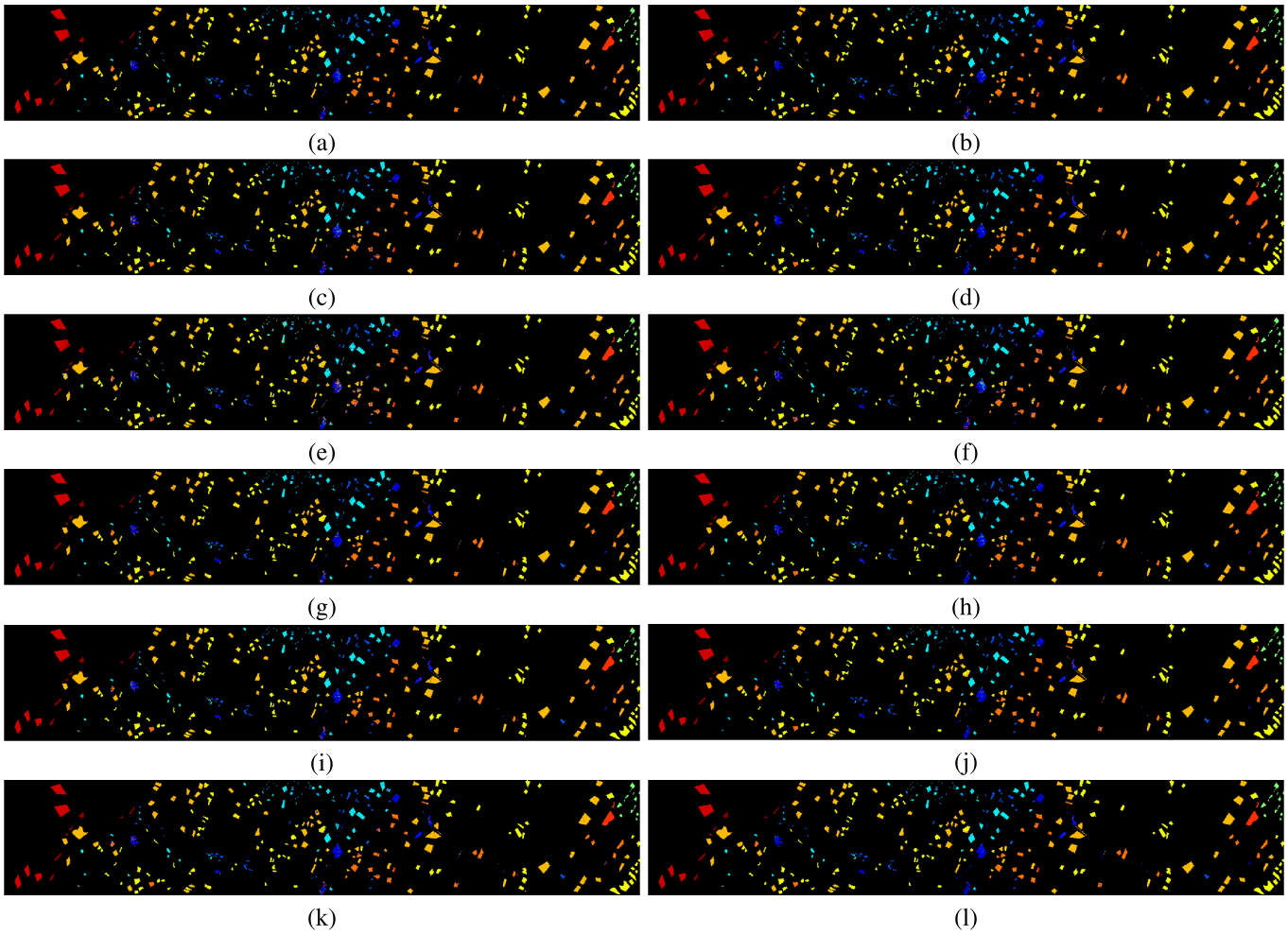


Fig. 15. Classification maps of different BS methods on the Dioni dataset with 27 bands by using SVM: (a) BS-Net, (b) MH-DRL, (c) CAE-UBS, (d) S-AEBS, (e) SRL-SOA, (f) RLFFC, (g) MSGCF, (h) S⁴P, (i) MR-Selection, (j) DISGT, (k) BSD-GCN, and (l) S²HGC.

TABLE VII
ABLATION ANALYSIS OF THREE DATASETS WITH MOA (%)

Methods	Indian Pines	Pavia University	Dioni
HGCBS	79.64	91.89	91.58
S²HGC-Random	81.76	92.34	92.17
S²HGC	**83.83**	**93.75**	**93.09**

correlations between bands more effectively. By seamlessly fusing spectral–spatial information, S²HGC exhibits robust adaptability to diverse datasets, making it a versatile and efficient solution for HSI application.

F. Visualization Analysis of S^2IF

To intuitively demonstrate the performance in extracting spectral and spatial features, superpixel-level features are remapped to the coordinate system of the original HSI. As shown in Fig. 16, the spatial response heatmaps correspond to three datasets generated by overlaying the features of Top5 bands selected by S²HGC. The regions indicated by the red boxes emphasize significant textures for different land-cover classes, reflecting their intrinsic relationship. Moreover, the spatial response of different objects exhibits strong separability, particularly at boundary and texture transition regions. It is revealed that the S²IF module effectively characterizes discriminative spatial information from different band combinations while reducing redundancy through multilevel information fusion. Furthermore, the collaborative fusion of spectral and spatial information by the S²IF module significantly enhances the ability to distinguish objects with similar spectral characteristics and distinct spatial structures. As illustrated in Fig. 9, the Top5 bands selected by S²HGC achieve high enough OA across all datasets, further validating the effectiveness of S²IF.

G. Analysis of Running Time

In this section, the computational efficiency of BS methods is analyzed. To ensure experimental fairness, all methods were executed under the same computing environment. The RLFFC, MSGCF, and S⁴P are executed in MATLAB and other methods are implemented in Python. Table VIII summarizes the running time to select 20 bands across three datasets. The results reveal that S²HGC demonstrates an obviously shorter running time compared to CNN-based, DRL-based, and graph-based BS methods, such as BS-Net, MH-DRL, MR-Selection, DISGT, and BSD-GCN, indicating its superior

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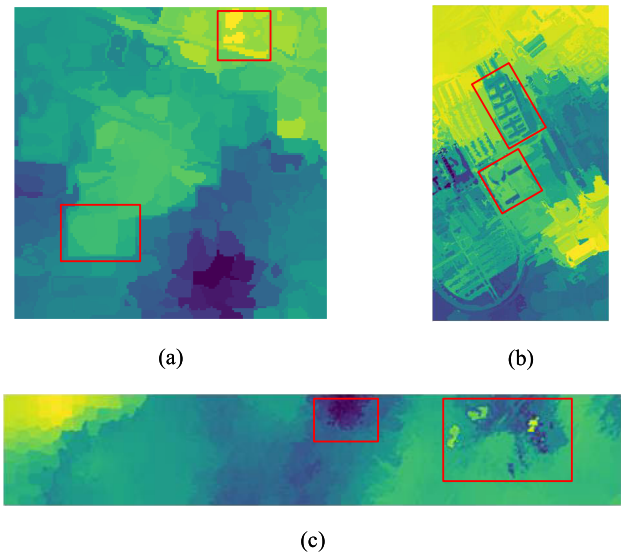


Fig. 16. Spatial response heatmaps output by the S^{2}IF module with the Top5 selected bands. (a) Indian Pines dataset. (b) Pavia University dataset. (c) Dioni dataset.

TABLE VIII
RUNNING TIME (S) FOR SELECTING 20 BANDS USING DIFFERENT METHODS

Methods	Indian Pines	Pavia University	Dioni
RLFFC	5.47	9.86	25.27
MSGCF	>1h	>1h	>1h
S^{4}P	3.26	20.53	36.26
BS-Net	364.86	806.49	1561.47
MH-DRL	424.03	913.56	2702.29
CAE-UBS	23.61	21.37	25.14
S-AEBS	56.48	49.37	54.83
SRL-SOA	33.87	32.38	31.18
MR-Selection	227.36	584.45	1035.67
DISGT	211.43	568.76	976.47
BSD-GCN	3457.84	>1h	>1h
S^{2}HGC	165.73	419.65	617.27

efficiency. In contrast, AE-based methods exhibit relatively shorter running time as they do not incorporate spatial information into the BS process. Moreover, RLFFC and S^{4}P also require less running time because they do not take much time to train the model. However, these methods generally achieve lower OA, indicating that while their computational speed is advantageous, the quality and effectiveness of the selected bands are compromised. In summary, S^{2}HGC strikes a balance between classification performance and computational efficiency, enabling the selection of crucial bands within an acceptable timeframe.

V. CONCLUSION

This article proposes S^{2}HGC, an unsupervised hyperspectral BS method that enhances information interaction across long-distance bands to effectively capture the nonlinear correlations between bands while optimizing the entire BS process within an end-to-end training framework. Notably, it is unnecessary for the preprocessing steps such as patch extraction or spectral calibration, thereby preserving the original spatial structure of HSIs. The CAE performs an initial partitioning of full bands into several combinations, effectively reducing redundancy. Moreover, the S^{2}IF module extracts significant spatial features from various band combinations and facilitates comprehensive S^{2}IF, which enables the model to achieve robust performance even within minimal or no labeled samples. Finally, an adaptive hypergraph convolution is employed to highlight higher-order dependencies among bands, overcoming the limitations of traditional graph structures that only model pairwise relationships and further enhancing long-distance information interaction through the introduction of transition nodes. Experimental results on publicly available HSI datasets demonstrate that S^{2}HGC derives superior performance, significantly improving classification performance compared to other state-of-the-art BS methods.

In summary, S^{2}HGC presents an innovative and effective approach to BS tasks, addressing the limitations of traditional methods that either rely solely on spectral information or focus primarily on local correlations between neighboring bands. Future research will focus on improving its applicability by exploring mechanisms for processing multiple datasets simultaneously and extending its potential to dataset adaptability and large-scale HSI datasets. At the same time, we will integrate domain knowledge and investigate broader application scenarios to further leverage the strengths of the proposed method in diverse real-world contexts, such as ecological function, land-use analysis, geological mapping, agricultural monitoring, and so on. In addition, similar to other BS methods, determining the optimal number of bands in the subset remains a significant challenge. Therefore, future efforts will aim to develop more advanced strategies to address these issues effectively.

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